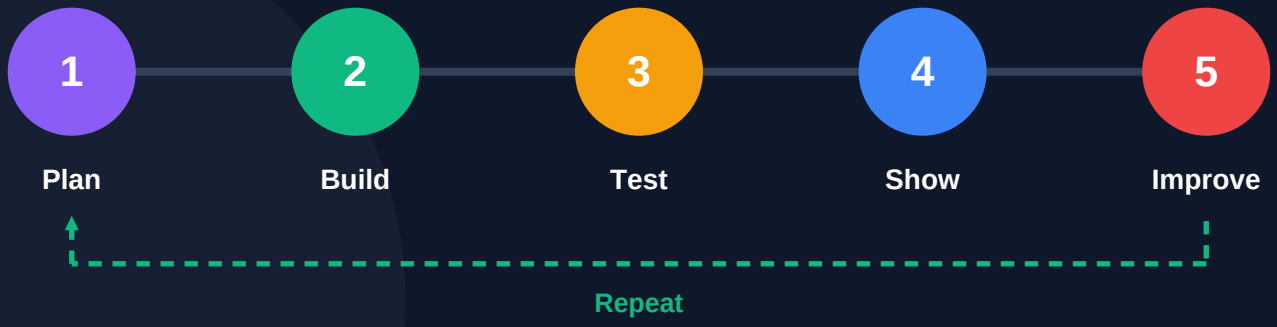
Agile Model

Explained from zero

with a real-time Facebook-like application example



Study guide for software testers

Agile is easy to understand if you imagine a company developing a Facebook-like application. Agile means **developing software little by little** instead of waiting until the entire application is finished.

Let's understand exactly what happens, who does the work, and how the process flows.

1 First, understand the basic idea

Imagine a customer says:

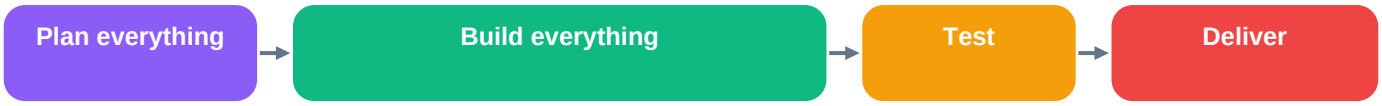
> "I want an application like Facebook."

The application needs

- Registration and login
- User profiles
- Photo uploads
- Friend requests
- Chat

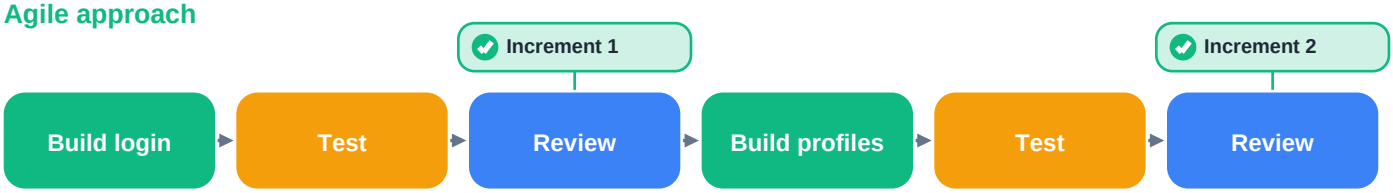
There are two ways to build it.

Traditional approach (Waterfall)



The customer sees working software only at the very end.

Agile approach



Software is delivered in small working pieces called increments.

Traditional approach (Waterfall): develop all the planned features, then complete the main testing and deliver the application. Agile approach: build a small set of features, test them, show the customer, and then continue with the next features.

> **The important difference:** Agile delivers software in small working pieces called **increments**.

2 How Agile works, step by step

Let’s imagine you are a software tester in this company.

1

Step 1: The customer explains the requirements

The customer tells the company what the application should do. For example:

“I want users to register, log in, upload photos, and connect with friends.”

The team discusses what the customer needs.

2

Step 2: The Product Owner creates a Product Backlog

A **Product Backlog** is a list of all the features and work the application may need.

Feature	Priority
Registration	High
Login	High
User profile	High
Photo upload	Medium
Friend requests	Medium
Chat	Later

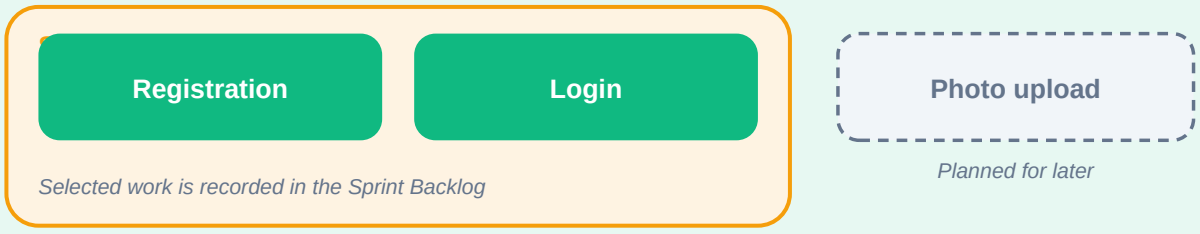
The Product Owner prioritizes the work. Priorities can change when the customer provides new information.

3

Step 3: The team selects work for a Sprint

A **Sprint** is a fixed period of work. In Scrum, it is one month or less, often two weeks.

Suppose the team decides to work on registration and login during Sprint 1. They do not try to finish the whole Facebook application in these two weeks. They focus on the selected work.



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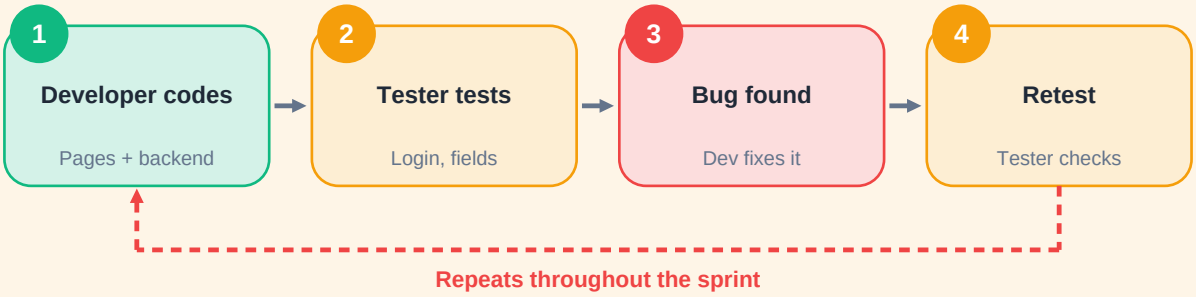
Step 4: Developers start coding and testers start testing

The team works together during the sprint.

- **Developer:** creates the registration page, login page, and backend logic.
- **Tester:** checks correct login, incorrect passwords, empty fields, and registration.
- **Developer fixes bugs:** if the tester finds a bug, the developer fixes it and the tester checks it again.



The tester does not necessarily wait until every feature has been coded before starting testing.

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Step 5: The team shows the working feature to the customer

At the end of Sprint 1, the team demonstrates registration and login. The customer can try the features and give feedback. For example:

"Registration and login work correctly. Now I also want users to reset forgotten passwords."

The Product Owner discusses the new request and decides its priority. It may be added to the Product Backlog.

6

Step 6: The team reviews how it worked

The team holds a **Sprint Retrospective**. They discuss questions such as:

- What went well?
- What problems occurred?
- How can we work better in the next sprint?

For example, the team decides to clarify requirements earlier so fewer bugs arise from misunderstandings.

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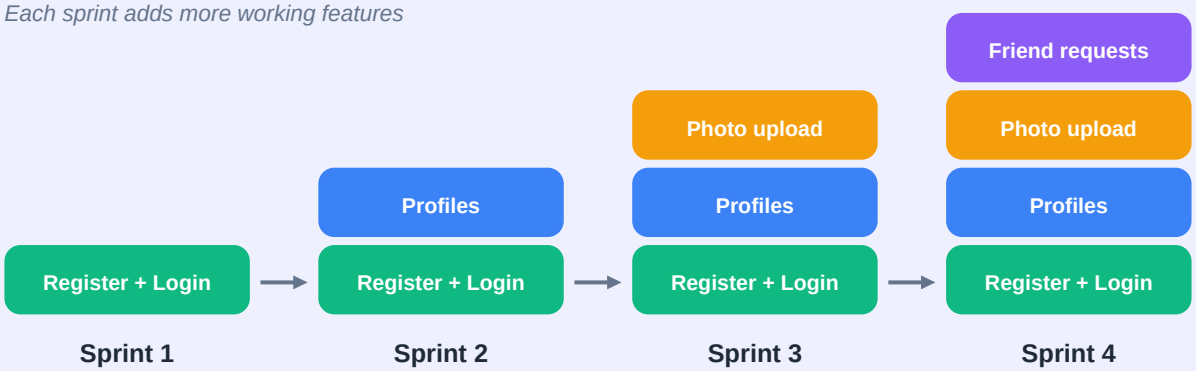
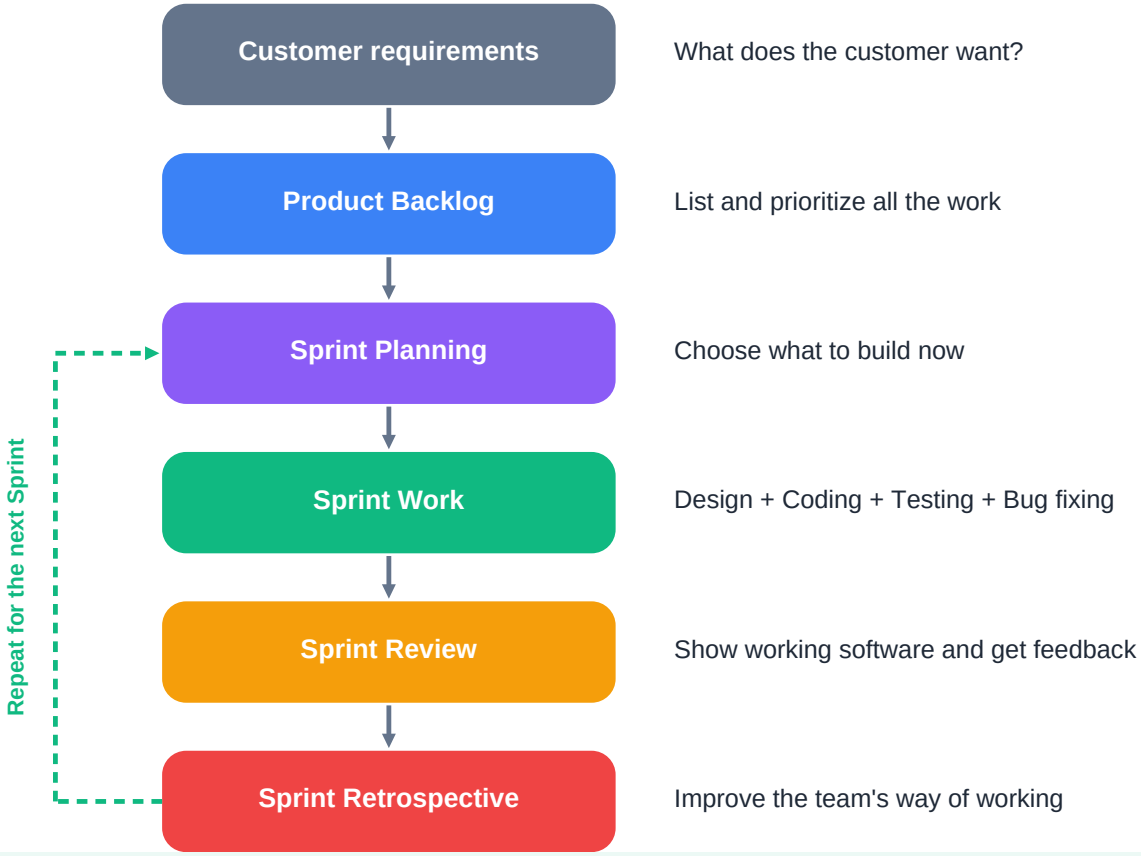
Step 7: The next Sprint begins

In Sprint 2, the team might work on user profiles. In Sprint 3, photo uploads. In Sprint 4, friend requests. Each sprint adds more working features to the application.

Each sprint adds more working features



3 The complete Agile flow



The product can be released after a sprint when it meets the release requirements. Teams do not always wait until every planned feature is finished to release useful software.

4 What is the tester’s job in Agile?

Suppose you are the tester assigned to the login feature.

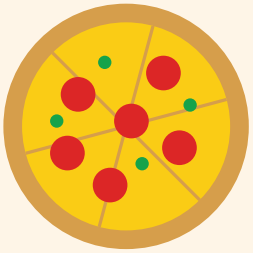
1. Read the login requirements.
2. Prepare test cases.
3. Test the login feature as soon as it is ready.
4. Report bugs in Jira.
5. Let the developer fix them.
6. Retest the fixes.
7. Check related features for regressions.
8. Share testing results with the team.

The goal: find problems early and help the team deliver working software during the sprint.

5 Remember these five things

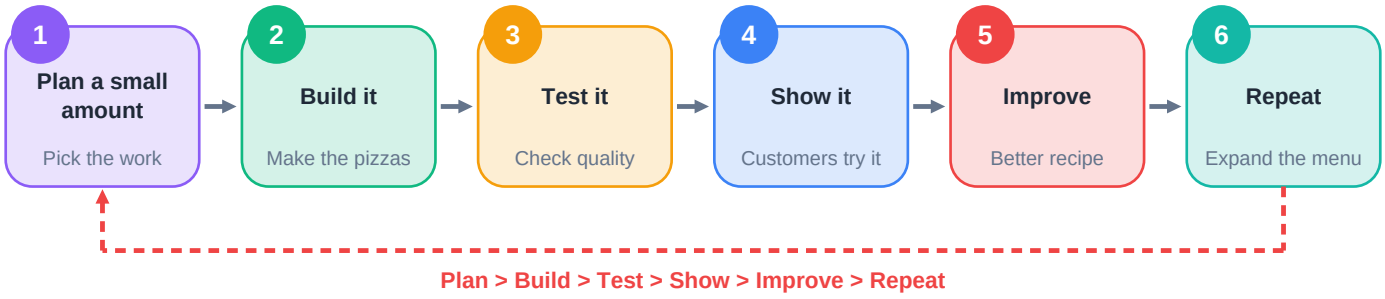
Term	Easy meaning
Product Backlog	All the work the product may need
Sprint	A fixed period of development
Sprint Backlog	Work selected for the current sprint
Sprint Review	Show the work and get feedback
Sprint Retrospective	Improve how the team works

The easiest way to remember Agile



Imagine making a **pizza business**. You prepare a few pizzas, let customers try them, collect feedback, improve the recipe, and then expand the menu.

Agile works similarly:



Plan > Build > Test > Show > Improve > Repeat